

# What is a software Agent

- **Carl Hewitt (1997)**

*“a self-contained, interactive and concurrently-executing object, with some encapsulated internal state and which could respond to messages from other similar objects”*

- **Gilbert, IBM**

*“software entity that carries out some set of operations on behalf of a user or another program with some degree of independence or autonomy, and in so doing, employ some knowledge or representation of the user’s goals or desires”*

# Software Agents (1)

- **An agent**
  - automates **frequent or tedious tasks**
  - determines **patterns in requests, or in the performance of its tasks**
  - frees the user **to accomplish other, more productive work**
  - adapts to the user **instead of the user having to adapt to a given application**
  - provides **some level of abstraction; determines what task is required, and the best way to execute it**
  - **presents a simple interface that hides the complexity of the system**
  - **should be able to resolve ambiguity and make decisions**
  - **should be able to learn from other agents the best manner in which to complete a given task**
  - **is intended to reduce waste of resources**

# Software Agents (2)

- **Weak characteristics**
  - **those characteristics of an agent that most experts agree the agent must possess in order to qualify as an agent**
  - **Autonomy is the trait that the agent is able to perform without the direct intervention of a human operator.**
  - **Social ability reflects that agents can act with other agents, and humans through some sort of interface and language.**
  - **Reactivity is the ability of agents to perceive the environment in which they find themselves, and to respond to changes in that environment.**
  - **Proactivity is the concept that agents are able to take the initiative to pursue a goal.**
  - **Learning means that an agent develops new knowledge from the task.**
  - **Temporal continuity reflects the fact that agents continually run (although they may go into a sleep state until needed).**
  - **Goal-oriented-ness is the characteristic of pursuing a goal, and of making decisions on how to go about completing complex tasks.**

# Software Agents (3)

- Strong characteristics
  - more specific traits that agents have; these, however, are not as generally accepted by researchers
  - Mobility is the ability of an agent to move about an electronic network.
  - Benevolence implies the idea that an agent is disposed to perform the tasks that it is asked to perform, and that it will not try to counter the wishes of its user.
  - Rationality is the concept that an agent will carry out its task in a logical manner, and with the “forethought” to avoid taking actions that will preclude its ability to carry out its goal, at least as much as possible, and as bound by its beliefs.
  - Adaptivity is the ability of an agent to adjust itself to the preferences, habits, and styles of its user.
  - Collaboration represents the idea that the agent will not blindly or unthinkingly attempt to perform the tasks of its user. Instead, it will ask questions, prompt for more information if needed, or refuse or postpone tasks that are impossible or dangerous. (For example, if a task will cause damage to other users of a system, the agent should refuse the task.)

# Software Agents (4)

- Other characteristics:
- Agency
  - the authority and autonomy possessed by an agent
  - its agency is enhanced by the degree to which it can represent the user
  - an advanced agent might manipulate data, services, or applications
  - a more advanced agent will collaborate and negotiate with other agents on the users' behalf
- Intelligence
  - the ability of the agent to reason and learn, as well as the agent's ability to understand the user's goals and desires, and to carry out its assigned tasks
  - the user might specify preferences in the form of a set of rules to aid in the understanding
  - at higher degrees of intelligence, an agent might possess the ability to infer or reason out the user's preference, resolve ambiguity, and to plan out means to achieve a goal
  - still higher levels of intelligence would include the ability to learn and adapt to an environment and develop relationships in the user's preferences, and in the resources available to the agent

# Agent Applications

- **workflow management**
- **network management**
- **air-traffic control**
- **smart databases**
- **command and control**
- **education**
- **data mining**
- **electronic commerce**
- **distributed systems monitoring and diagnosis, test and repair**
- **.....**



# Agent communication

- **At software level**
  - FIPA (Foundation for Intelligent Physical Agents) ACL (Agent Communication Language)
  - FIPA ACL specs describe aspects of the structure of messages and the ontology service
  - Usable: IIOP (Internet Inter-ORB Protocol), WAP (Wireless Application Protocol), HTTP (HyperText Transfer Protocol), TCP/IP over wire/wireless, bit-oriented, string-oriented and XML-oriented message representations
  - Our agents currently use TCP/IP sockets and XML
- **At hardware level**
  - Serial, I2C, Ethernet or other, whatever is available
  - Our system currently uses leased phone lines, Ethernet cable and Wi-Fi
  - Under development: encrypted communication through GPRS/EDGE/EV-DO



# Agent Platforms – ADK

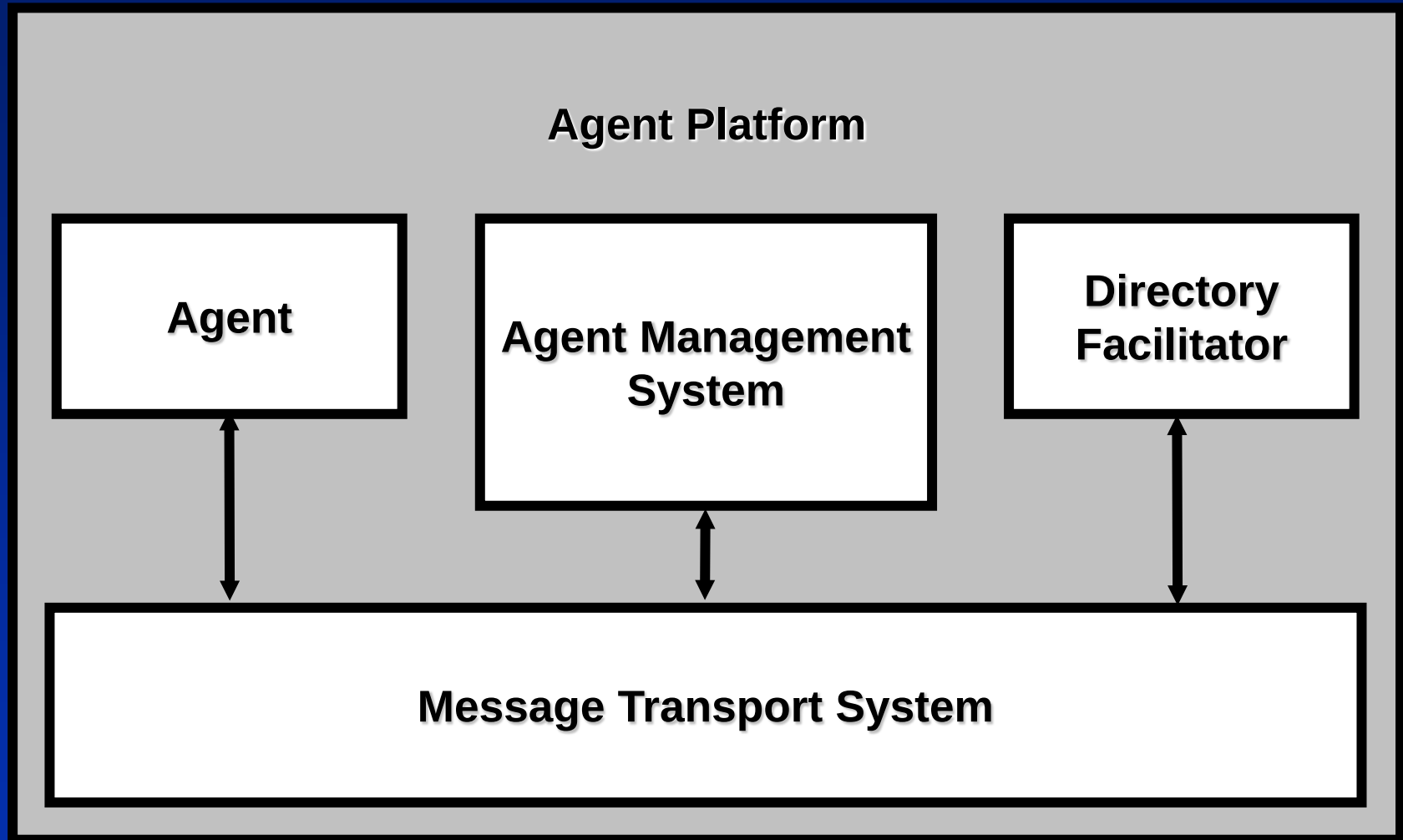
- **ADK – Agent Development Kit is 100% Java**
- **Strong mobility (data, code and execution state are transferred along with the agent)**
- **Typical for ADK:**
  - **Dispatcher performs load-balancing**
  - **The Connector agent is the interface between the society and the database**
  - **The Wrapper provides uniform access to various hardware**



# Agent Platforms – JADE

- **JADE - Java Agent DEvelopment framework is 100% Java**
- **Adaptive message transport mechanism**
- **Agent Management System (AMS) is the authority across all machines**
- **Each agent has a database of some diagnosis info for verifying several modules**
- **If the agent has the diagnosis info it then simply calls for the diagnosis functions to start the check**
- **If diagnosis info is unavailable then it asks the other agents within the society for it**

# JADE Platform





# Other Agent Platforms

- **JADE – LEAP (Light Extensible Agent Platform )**
  - an extension of JADE
  - developed in MIDP (Sun's Java Micro Edition) and pJava
- **AgentLight (The Platform for Lightweight Agents)**
  - inference engine is based on a Prolog-like First-Order-Logic
  - a container is compiled into a single executable and deployed to the portable device